

The background of the slide is a photograph of an industrial facility, likely a power plant or refinery, with several tall smokestacks emitting thick plumes of white smoke. The scene is set against a dramatic sky at sunset or sunrise, with warm orange and yellow light illuminating the clouds and the smoke. The overall tone is somewhat somber due to the industrial nature of the scene, but the lighting adds a sense of atmospheric depth.

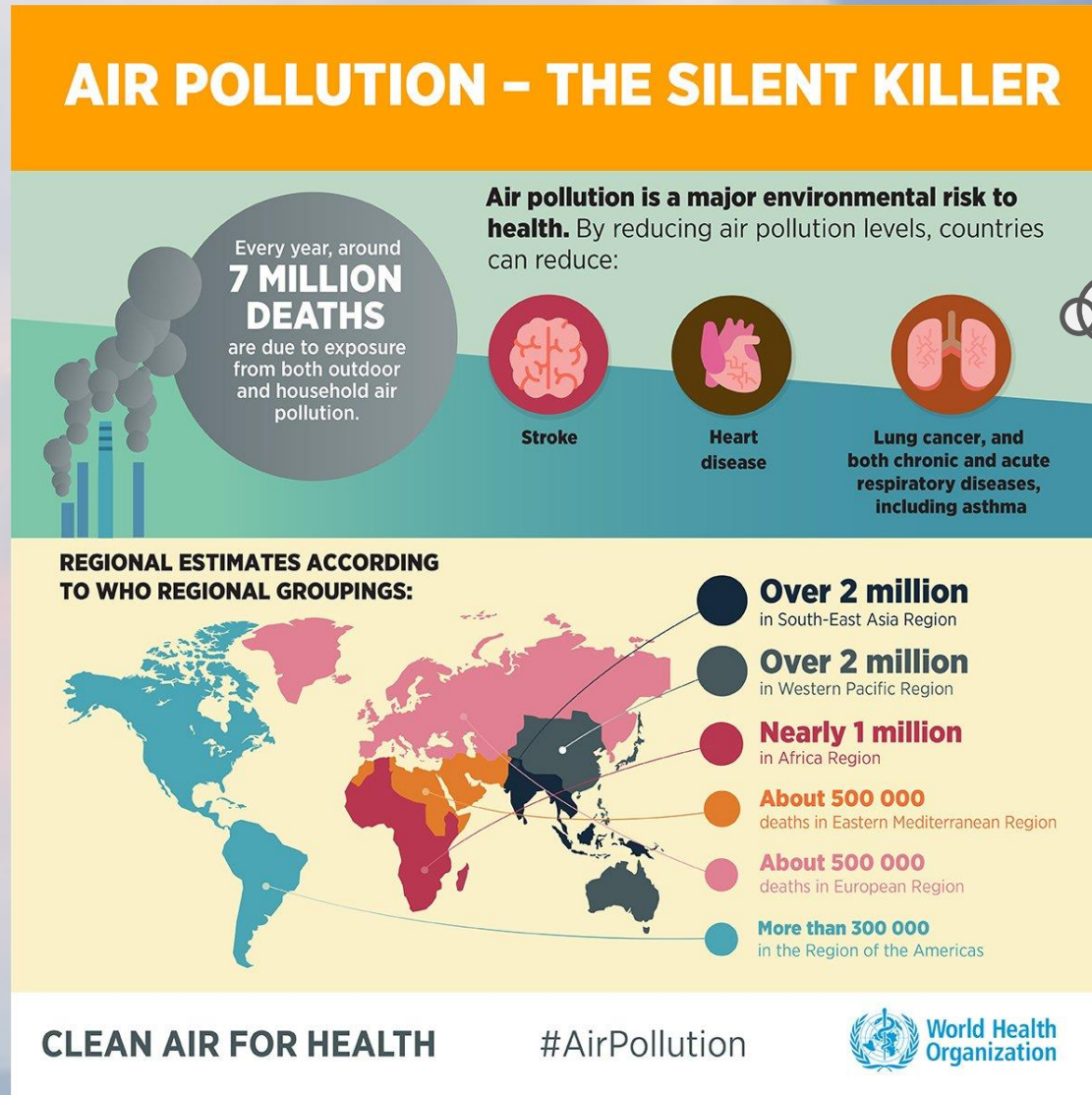
Graph Neural Network For Air Quality Forecasting

基於圖神經網路之空氣品質預測

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Effects of Air Pollution

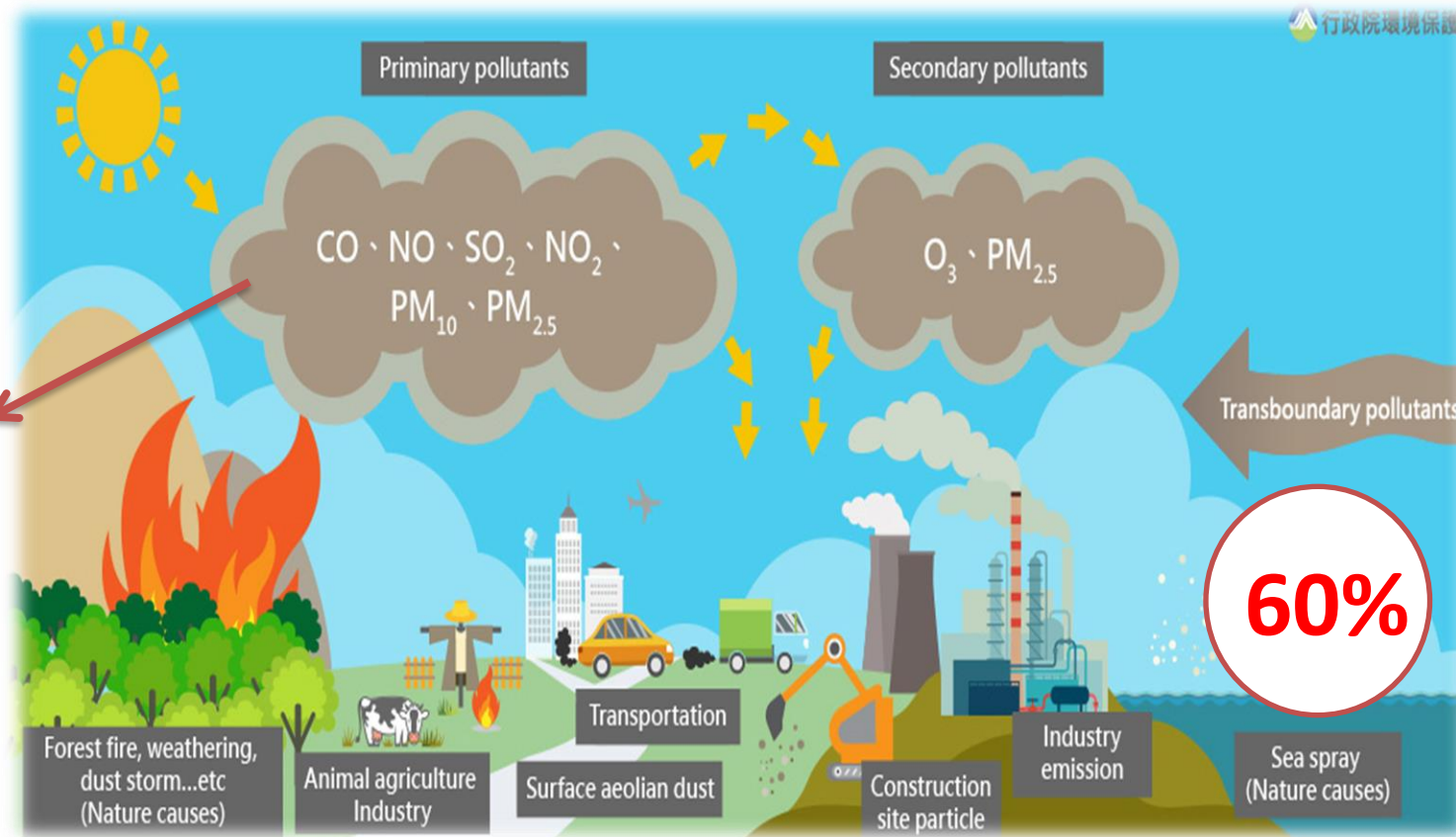


28% Heart disease and stroke
43% Lung disease
29% Lung cancer

Air pollution causes **20,000 people early deaths** each year in Taiwan.

Definition of AQI

行政院環境保護署



Based on the measurement of those solid and liquid particles.

Air Quality Index (AQI)

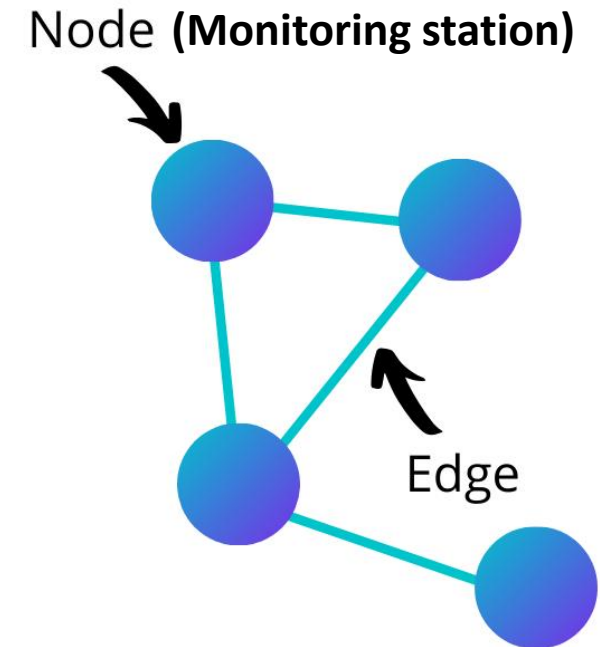
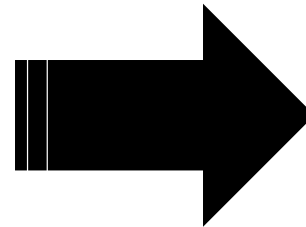
Good	Moderate	Unhealthy for Sensitive Groups	Unhealthy	Very Unhealthy	Hazardous
0~50	51~100	101~150	151~200	201~300	301~500

Air Quality Prediction

- ✓ Why do we need GNN?
 - To process non-Euclidean data!



Complex Air Quality Monitoring Network



■ Air pollutant (Hourly time series data)

→ Affected by many factors (such as weather, traffic volume).

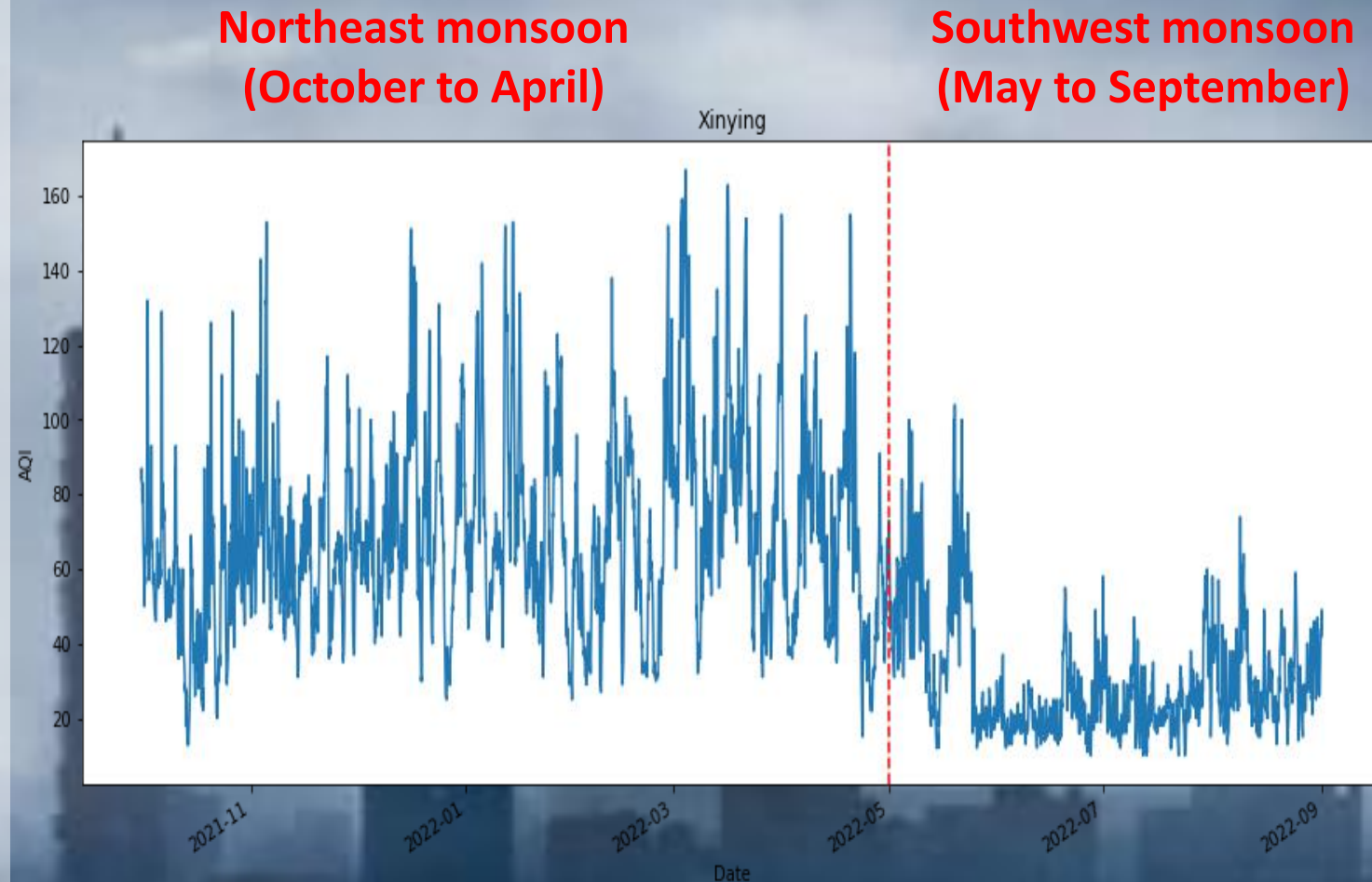
◆ How Weather Affects Air Quality?

The two main factors:

(1) Wind direction

(2) Terrain obstruction

➤ Stronger winds -> Lower air pollution concentrations



Nearly a year of Tainan air-quality distribution

Research methods

■ Data Preprocessing



Environmental Protection Administration
Executive Yuan, R.O.C. (Taiwan)

- Dataset: Air quality index (AQI) (historical data, updated hourly)



Data
filtering



Tainan

- 6 air monitoring stations

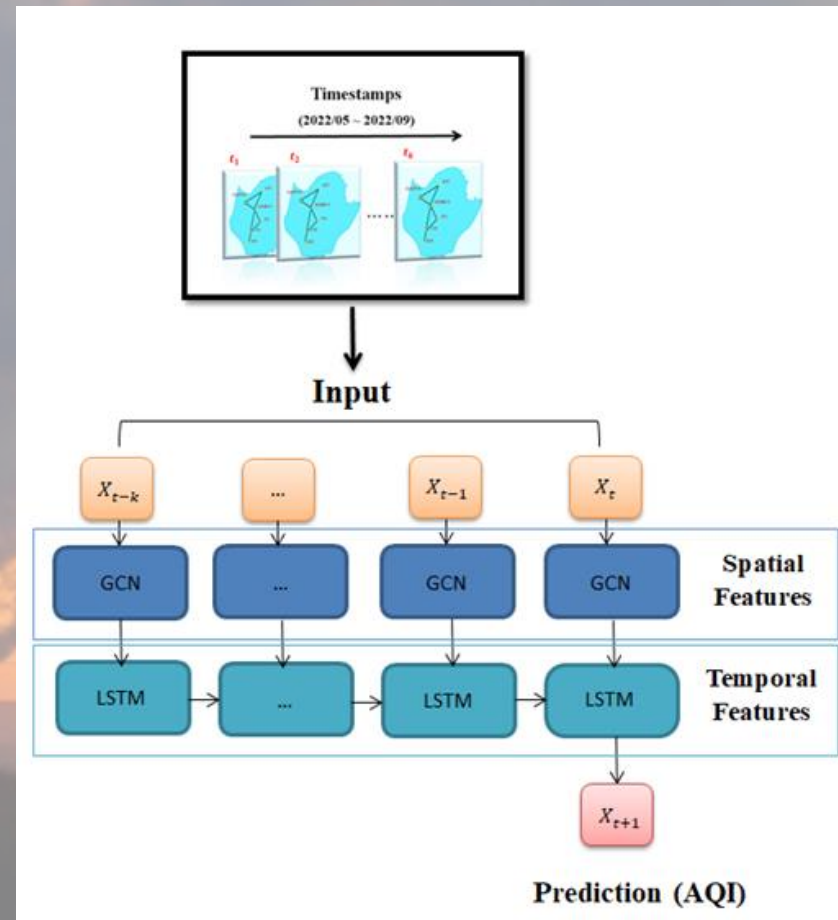
Filling
missing
value



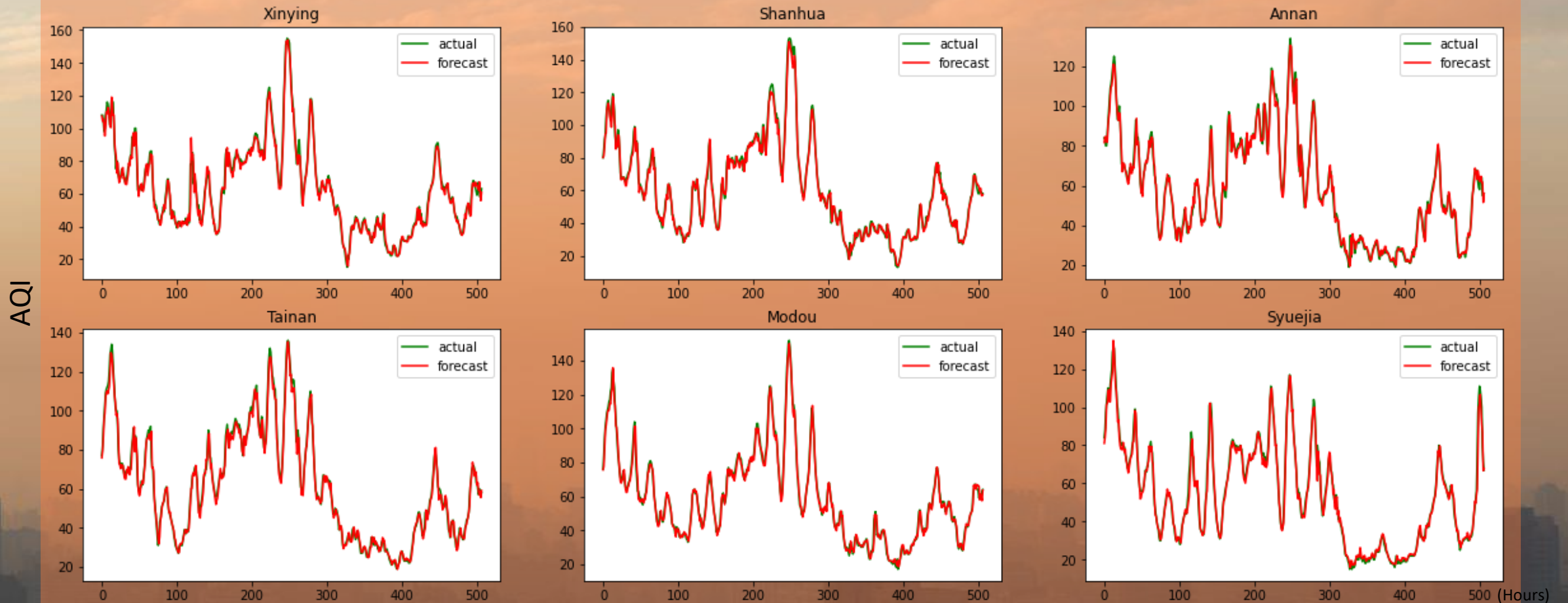
Splitting
data

- Southwest/Northeast monsoon dataset

Model



Results



Dataset: 2021/10/01 00:00 ~ 2022/04/30 23:00 (Northeast monsoon)

(Using the **past 3 hours** of data to predict the **next 1 hour**)

Results

(Tainan)



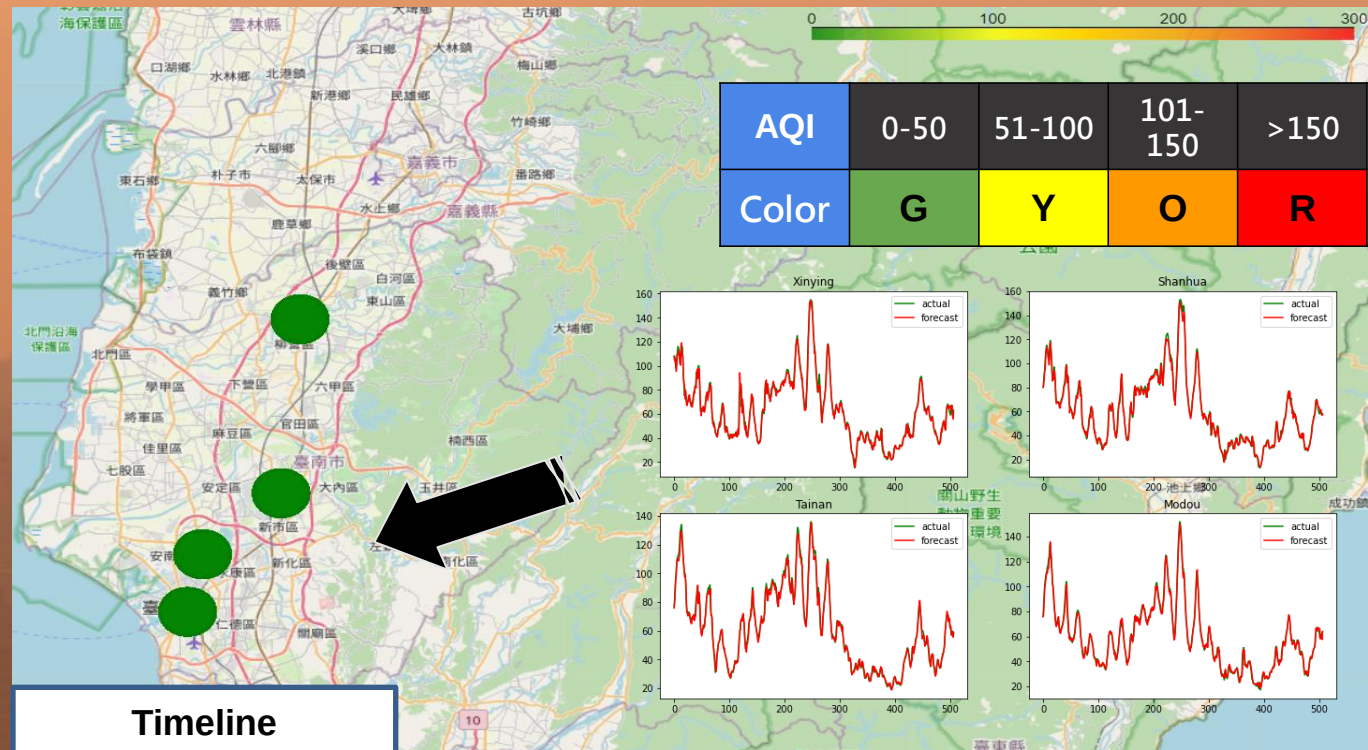
LSTM 3hr	RMSE	MAE	MAPE (%)
1hr later	3.8938	2.7056	4.6262
2hr later	6.9457	4.9455	8.3244
3hr later	9.6321	6.9617	12.6157

GCN-LSTM 3hr	RMSE	MAE	MAPE (%)
1hr later	3.1903	2.3181	4.4209
2hr later	5.7727	4.2287	7.7201
3hr later	8.5973	6.4668	12.0718

- The results of this study evaluated both models using root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE).

Future works

1. Add more nodes and features: Highly correlated with air quality, such as PM2.5, O3, wind direction, etc.
2. Data visualization: Using PHP and MySQL tools to present website information in the WordPress framework and apply it to the Leaflet map.





Thank You For Listening